Open problems:
Non-linear representations in quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Madelung representation.** *Research.* "The Madelung equations ... provide such a representation", one "not even tied to vector spaces". The equivalence of the Madelung equations with quantum theory, and in what sense they represent QST:H, are not shown. As stated the equivalence is false: the Madelung equations have solutions that come from no wave function, and to exclude them one must add a quantization condition on the circulation of the velocity field, $\oint \nabla S \cdot d\ell \in 2\pi\hbar\,\mathbb{Z}$, which is not part of the hydrodynamic equations and is hard to state at the nodes, where the density vanishes and the phase S is undefined (T. C. Wallstrom, Phys. Rev. A **49** (1994), 1613). A precise statement is needed first, and checking it needs partial differential equations on $\mathbb{R}^3$ with a singular nodal set, which neither Mathlib nor TauCeti provides. (Brief, Sec. 5.)

From the formalization

1. None found. The blocker of the previous version of the brief (Theorem 1 as an independence statement) is resolved: the revised brief states its mathematical content as Proposition 1 and Corollary 3, which are checked in Lean, and argues Theorem 4 from them informally, which is checked by reading (Log, "Differences from the brief").